

List of Presentation and Talks

Oliver Gutsche

May 8, 2026

O. Gutsche,

Scientific Computing and Scientific Software Infrastructure - High Performance Computing (HPC), (2025), Presentation for the [Chicagoland Computational Traineeship in High Energy Particle Physics](#), ([Material](#))

O. Gutsche,

Scientific Computing and Scientific Software Infrastructure - Particle Physics Overview, (2025), Presentation for the [Chicagoland Computational Traineeship in High Energy Particle Physics](#), ([Material](#))

O. Gutsche,

Computing and Software Infrastructure, (2024), Presentation at the [Fermilab-CERN HCP Summer School 2024](#), ([Material](#))

O. Gutsche,

HL-LHC Computing, (2023), Presentation at the [USCMS Undergraduate Summer Internship 2023](#), ([Material](#))

O. Gutsche,

The U.S. CMS HL-LHC R&D Strategic Plan, (2023), Parallel Session Talk at the [26th International Conference on Computing in High Energy & Nuclear Physics \(CHEP2023\)](#), ([Material](#))

O. Gutsche,

Computing, (2022), Plenary talk given at the DOE/HEP Review of the Energy Frontier Laboratory Research Program, ([Material available upon request](#))

O. Gutsche,

Computing, (2022), Lecture given at [17th Hadron Collider Physics Summer School](#), ([Material](#))

O. Gutsche,

Exabytes and the Large Hadron Collider: the importance of computing excellence for scientific discovery, (2019), Seminar given at [Dr. Leonard Weisenthal Colloquium Series at Lewis University](#), ([Material](#))

O. Gutsche,

WLCG Web Proxy Auto Discovery for Dynamically Created Web Proxies, (2019), Parallel Session Talk at the [24th International Conference on Computing in High Energy & Nuclear Physics \(CHEP2019\)](#), ([Material](#))

O. Gutsche,

Bringing heterogeneity to the CMS software framework, (2019), Parallel Session Talk at the [24th International Conference on Computing in High Energy & Nuclear Physics \(CHEP2019\)](#), ([Material](#))

O. Gutsche,

Highlights from Parallel Session: Software & Computing, (2019), Plenary Talk at the [2019 US CMS Annual Collaboration Meeting](#), ([Material](#))

O. Gutsche,

Software & Computing, (2019), Plenary Talk at the [2019 US CMS Annual Collaboration Meeting](#), ([Material](#))

O. Gutsche,

CMS Software, Computing and Analysis Tools, (2019), Plenary Talk at the CMS Data Analysis School 2019 ([CMSDAS 2019](#)), ([Material](#))

O. Gutsche,

Exploring the Energy Frontier using Amazon Web Services, (2018), Amazon Booth at [Supercomputing 2018](#) in Dallas, TX, ([Material](#))

O. Gutsche,

The Future of Large Scale Scientific Computing - Trends in computing hardware, (2018), [US ATLAS Workshop 2018 Open Day](#) at the University of Pittsburgh, ([Material](#))

O. Gutsche,

The Future of Large Scale Scientific Computing, (2018), [Colloquium](#) given at the Physics Department at the University of Massachusetts Amherst, ([Material](#))

O. Gutsche,

CMS Software, Computing and Analysis Tools, (2018), Plenary Talk at the CMS Data Analysis School 2018 ([CMSDAS 2018](#)), ([Material](#))

O. Gutsche,

Data Organisation, Management and Access (DOMA) in High Energy Physics, (2017), Talk at the Workshop for Data Organisation, Management and Access (DOMA) in Astronomy, Genomics and High Energy Physics ([DOMA2017](#)), ([Material](#))

O. Gutsche,

The Future of Large Scale Scientific Computing, (2017), [Colloquium](#) given at the Department of Physics and Astronomy at Texas Tech University, ([Material](#))

O. Gutsche,

Particle Physics - A world wide journey from recording particles to analysis using big computing, (2017), Plenary Talk at the Chicago Council on Science & Technology Panel: [Fermilab and the New Frontiers of Physics](#), ([Material](#))

O. Gutsche,

CMS Analysis and Data Reduction with Apache Spark, (2017), Parallel Session Talk at the 18th International Workshop on Advanced Computing and Analysis Techniques in Physics Research ([ACAT 2017](#)), ([Material](#))

O. Gutsche,

50 years Fermilab - Computing Innovations, (2017), Talk at the Fermilab 50th Anniversary [Symposium](#), ([Material](#))

O. Gutsche,

Status and Plans of the CMS Big Data Project, (2017), Talk at the [CERN Database Futures Workshop 2017](#), ([Material](#))

O. Gutsche,

Data Management, Access and Organisation (DOMA), (2017), Summary Talk at the [2nd S2I2 HEP/CS Workshop](#), ([Material](#))

O. Gutsche,

Data Analytics in Physics Data Reduction, (2017), Talk at the [CERN openlab Workshop on Machine Learning and Data Analytics 2017](#), ([Material](#))

O. Gutsche,

Panel Discussion on the S2I2 Conceptualization for HL-LHC, (2017), Talk at the Openscience Grid All Hands Meeting ([OSG AHM 2017](#)), ([Material](#))

O. Gutsche,

The Fermilab HEPCloud, or How to add 240 TFlops in an Hour or Two, (2017), Talk at HEP Software Foundation [Workshop](#), ([Material](#))

O. Gutsche,

Intel Big Data Project: CMS Physics Data Reduction Use Case, (2016), Talk at the [CERN openlab Technical Workshop 2016](#), ([Material](#))

O. Gutsche,

Big Data in HEP: a Comprehensive Use Case Study, (2016), Parallel Session Talk at 22nd International Conference on Computing in High Energy and Nuclear Physics ([CHEP 2016](#)), ([Material](#))

O. Gutsche,

Big Data at FNAL, (2016), Plenary Talk given at the [8th INFIERI Workshop](#), ([Material](#))

O. Gutsche,

Amazon and the Future of Computing, (2016), [Ask-A-Scientist](#) Lecture given to the general public at Fermi National Accelerator Laboratory, ([Material](#))

O. Gutsche,

Exascale and Exabytes: Future directions in HEP software and computing, (2015), Seminar given at [Center of Research Computing Seminar Series](#) the University of Notre Dame, ([Material](#))

O. Gutsche,

Exascale and Exabytes - Computing at Fermilab for LHC and Intensity Frontier, (2015), Seminar given at the University Los Andes in Bogota, Colombia, ([Material](#))

O. Gutsche,

Future directions in HEP Software and Computing, (2015), Plenary talk given at the Meeting of the Division of Particles & Fields of the American Physical Society at the University of Michigan ([DPF2015](#)), ([Material](#))

O. Gutsche,

Diversity in Computing Technologies and Strategies for Dynamic Resource Allocation, (2015), Plenary talk given at the Conference for Computing in High Energy Physics ([CHEP2015](#)), Okinawa, Japan , ([Material](#))

O. Gutsche,

CMS Computing Operations: Outlook on LHC Run 2, (2015), Talk given at the Openscience Grid All Hands Meeting 2015 at Northwestern University ([OSGAHM2015](#)), ([Material](#))

O. Gutsche,

Computing Facilities and Services, (2015), Talk given at the [Fermilab Institutional Review 2015](#), ([Material](#))

O. Gutsche,

Scientific Computing, (2014), Plenary talk given at the [47th Annual Fermilab Users Meeting](#), ([Material](#))

O. Gutsche,

CMS computing operations: the year ahead, (2014), Talk given at the Openscience Grid All Hands Meeting 2014 at SLAC ([OSGAHM2014](#)), ([Material](#))

O. Gutsche,

Recent Top Pair Asymmetry Measurements at CMS, (2014), Seminar given at the [Fermilab Joint Experimental-Theoretical Physics Seminar](#), ([Material](#))

O. Gutsche,

Computing: From Data to Physics, (2013), [Ask-A-Scientist](#) Lecture given to the general public at Fermi National Accelerator Laboratory, ([Material](#))

O. Gutsche,

Measurement of top quark properties with CMS, (2013), Parallel session talk given at the 19th International Symposium on Particles, Strings, and Cosmology ([PASCOS2013](#)), Taipei, Taiwan, ([Material](#))

O. Gutsche,

CMS: T2 dynamic data placement and plans for T2 disk space handling, (2013), Plenary talk given at the [2013 WLCG Collaboration Workshop](#) in Copenhagen, Denmark, ([Material](#))

O. Gutsche,

CMS: Disk/Tape Separation, (2013), Plenary talk given at the [2013 WLCG Collaboration Workshop](#) in Copenhagen, Denmark, ([Material](#))

O. Gutsche,

Computing Support Facilities, (2013), Talk given at the [2013 Fermi National Accelerator Laboratory Annual Science and Technology Review](#), ([Material](#))

O. Gutsche,

CMS computing operations in LHC run 1, (2013), Parallel session talk given at the 2013 Conference for Computing in High Energy Physics ([CHEP2013](#)), Amsterdam, The Netherlands, ([Material](#))

O. Gutsche,

CMS Production Transfer Case Study, (2013), Talk given at the [2013 Department of Energy ESNet Requirements Review](#), ([Material](#))

O. Gutsche,

Summary of CMS Operations Procedures, (2012), Talk given at the [WLCG Collaboration Workshop 2012](#), ([Material](#))

O. Gutsche,

Summary of Event Processing Track, (2010), Plenary talk given at the conference for Computing in High Energy and Nuclear Physics 2010 ([CHEP2010](#)) in Taipei (Taiwan), ([Material](#))

O. Gutsche,

Validation of software releases for CMS, (2009), Parallel session talk given at the conference for Computing in High Energy and Nuclear Physics 2009 ([CHEP2009](#)) in Prague, Czech Republic, ([Material](#))

O. Gutsche,

Dilepton Final States Studies at CMS: From the Standard Model to New Physics, (2009), Talk given at the [LHC and Dark Matter Workshop](#) at the University of Michigan in Ann Arbor, MI, ([Material](#))

O. Gutsche,

WLCG scale testing during CMS data challenges, (2007), Parallel session talk given at the conference for Computing in High Energy and Nuclear Physics 2007 ([CHEP2007](#)) in Victoria, Canada, ([Material](#))

O. Gutsche,

Distributed CMS Analysis on the Open Science Grid, (2006), Parallel session talk given at the conference for Computing in High Energy and Nuclear Physics 2006 ([CHEP2006](#)) in Mumbai, India, ([Material](#))